

Diffusion and Flow Matching as Implicit Jacobian Regularizers for Low-Variance Conditional Functions

Theory note draft

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Abstract

We study conditional generative modeling in the low-variance regime, where the target distribution is concentrated near the graph of an unknown function $f^* : c \mapsto y$. Although diffusion models and flow-matching models are usually motivated as distributional generative models, we show that, under Lipschitz constraints on the learned score or velocity field, they can induce a stricter Jacobian bound on the recovered clean conditional map than direct regression with the same Lipschitz budget. The key observation is that the clean prediction $f(c)$ is not represented directly; it is reconstructed from a score or velocity field through a schedule-dependent coefficient. For score-based diffusion, this coefficient is approximately σ_t^2/α_t . For rectified flow, it is $1 - t$. For trigonometric variance-preserving flow paths, it is $(2/\pi) \cos(\pi t/2)$. Whenever this coefficient is less than one, the implied clean predictor has a smaller conditioning-Jacobian bound than a directly parameterized regressor with comparable Lipschitzness. This provides a possible explanation for why diffusion and flow-matching objectives can act as additional regularizers for nearly deterministic conditional prediction, especially for near-support out-of-distribution inputs. The result does not imply semantic correctness far outside the training support; rather, it states that among possible extensions, Lipschitz-constrained diffusion and flow fields favor lower-Jacobian extensions of the conditional mean or mode.

1 Introduction

Consider the conditional learning problem

$$y = f^*(c) + \xi, \tag{1}$$

where c is a conditioning input and the noise ξ has small variance. In this regime, the conditional distribution $p(y \mid c)$ is concentrated near the graph of a deterministic function. Standard regression learns a map

$$g_\theta(c) \approx f^*(c). \tag{2}$$

Conditional diffusion and flow-matching models instead learn a time-indexed denoising field, score field, or velocity field around the graph of f^* .

The central claim of this note is the following:

In low-variance conditional prediction, diffusion and flow matching can implicitly regularize the Jacobian of the recovered clean map $c \mapsto f(c)$, including for near-support out-of-distribution inputs, provided the learned score or velocity network has bounded Lipschitzness in the conditioning input.

This is not a claim that diffusion identifies the correct $f^*(c)$ outside the data support. Out-of-support extrapolation remains statistically underdetermined. The claim is more modest: under equal Lipschitz budgets, the diffusion/flow representation can impose a smaller Jacobian bound on the implied clean predictor than direct regression.

The intuition is simple. A regressor directly outputs $f_\theta(c)$, so its Jacobian is bounded by the network Lipschitz constant. A diffusion or flow model outputs a field s_θ or v_θ , from which the clean $f_\theta(c)$ is reconstructed through a schedule-dependent coefficient κ_t . If the field is L -Lipschitz in c , the implied clean map is only $\kappa_t L$ -Lipschitz. When $\kappa_t < 1$, this gives a stricter Jacobian bound.

Contributions. This note makes three points.

1. We formalize low-variance conditional learning as learning a thin conditional density tube around the graph of f^* .
2. We derive Jacobian bounds for clean predictors reconstructed from score, noise-prediction, and flow-matching parameterizations.
3. We analyze the schedule-dependent coefficient κ_t for common diffusion and flow-matching schedules and compare them to direct regression.

2 Setup

Let $c \in \mathcal{C} \subset \mathbb{R}^{d_c}$ be a conditioning variable and $y \in \mathbb{R}^{d_y}$ be the output. Assume the data distribution satisfies

$$Y = f^*(C) + \xi, \quad (3)$$

where

$$\mathbb{E}[\xi \mid C = c] = 0, \quad \text{Cov}(\xi \mid C = c) \preceq \tau^2 I. \quad (4)$$

We focus on the low-variance regime $\tau^2 \ll 1$, where $p(y \mid c)$ is concentrated near the graph

$$\mathcal{M} = \{(c, y) : y = f^*(c)\}. \quad (5)$$

Let $\mathcal{S} = \text{supp}(p_C)$ denote the support of the observed conditioning distribution. We define the near-support OOD region by

$$\mathcal{S}_r = \{c \in \mathcal{C} : \text{dist}(c, \mathcal{S}) \leq r\}. \quad (6)$$

The analysis concerns regularization on $\mathcal{S}_r$, not arbitrary far-ODD extrapolation.

2.1 Direct regression

A regression model directly predicts

$$g_\theta(c) \approx f^*(c). \quad (7)$$

Assume g_θ is L -Lipschitz in c . Then almost everywhere,

$$\|J_c g_\theta(c)\|_{\text{op}} \leq L. \quad (8)$$

2.2 Conditional diffusion or flow fields

A diffusion or flow model learns a field

$$F_\theta(y_t, c, t), \quad (9)$$

where F_θ may be a score, noise predictor, velocity field, or related parameterization. Assume that for each fixed (y_t, t) , the field is also L -Lipschitz in c :

$$\|F_\theta(y_t, c, t) - F_\theta(y_t, c', t)\| \leq L\|c - c'\|. \quad (10)$$

The clean predictor is reconstructed from the field:

$$\hat{f}_{\theta,t}(c; y_t) = R_t(y_t, F_\theta(y_t, c, t)). \quad (11)$$

The key question is whether

$$\|J_c \hat{f}_{\theta,t}(c; y_t)\|_{\text{op}} \quad (12)$$

can be bounded by a smaller constant than L .

3 Score-Based Diffusion: The Basic Jacobian Shrinkage Lemma

Consider a Gaussian corruption process

$$Y_t = \alpha_t Y + \sigma_t Z, \quad Z \sim \mathcal{N}(0, I). \quad (13)$$

In the low-variance regime,

$$Y \approx f^*(c). \quad (14)$$

Then

$$Y_t \mid c \approx \mathcal{N}(\alpha_t f^*(c), \sigma_t^2 I). \quad (15)$$

The conditional score is

$$s_t^*(y_t, c) = \nabla_{y_t} \log p_t(y_t \mid c) = -\frac{y_t - \alpha_t f^*(c)}{\sigma_t^2}. \quad (16)$$

Solving this equation for $f^*(c)$ gives

$$f^*(c) = \frac{1}{\alpha_t} (y_t + \sigma_t^2 s_t^*(y_t, c)). \quad (17)$$

Thus, given a learned score model s_θ , define the implied clean predictor

$$\hat{f}_{\theta,t}(c; y_t) = \frac{1}{\alpha_t} (y_t + \sigma_t^2 s_\theta(y_t, c, t)). \quad (18)$$

Differentiating with respect to c while holding y_t fixed yields

$$J_c \hat{f}_{\theta,t}(c; y_t) = \frac{\sigma_t^2}{\alpha_t} J_c s_\theta(y_t, c, t). \quad (19)$$

Therefore, if

$$\|J_c s_\theta(y_t, c, t)\|_{\text{op}} \leq L, \quad (20)$$

then

$$\boxed{\|J_c \hat{f}_{\theta,t}(c; y_t)\|_{\text{op}} \leq \frac{\sigma_t^2}{\alpha_t} L.} \quad (21)$$

If the data noise is nonzero but small, with covariance approximately $\tau^2 I$, then

$$Y_t \mid c \approx \mathcal{N}(\alpha_t f^*(c), (\sigma_t^2 + \alpha_t^2 \tau^2) I), \quad (22)$$

and the coefficient becomes

$$\kappa_t^{\text{score}} = \frac{\sigma_t^2 + \alpha_t^2 \tau^2}{\alpha_t}. \quad (23)$$

In the low-variance limit $\tau^2 \rightarrow 0$,

$$\kappa_t^{\text{score}} \rightarrow \frac{\sigma_t^2}{\alpha_t}. \quad (24)$$

Theorem 1 (Diffusion-induced Jacobian regularization). *Assume:*

1. $Y = f^*(C) + \xi$, with $\text{Cov}(\xi \mid C = c) \preceq \tau^2 I$.
2. The score model $s_\theta(y_t, c, t)$ is L -Lipschitz in c , uniformly over the near-support region $\mathcal{S}_r$.
3. The implied clean map is reconstructed as

$$\hat{f}_{\theta,t}(c; y_t) = \frac{1}{\alpha_t} (y_t + (\sigma_t^2 + \alpha_t^2 \tau^2) s_\theta(y_t, c, t)). \quad (25)$$

Then for all differentiability points $c \in \mathcal{S}_r$,

$$\|J_c \hat{f}_{\theta,t}(c; y_t)\|_{\text{op}} \leq \frac{\sigma_t^2 + \alpha_t^2 \tau^2}{\alpha_t} L. \quad (26)$$

In particular, if

$$\frac{\sigma_t^2 + \alpha_t^2 \tau^2}{\alpha_t} < 1, \quad (27)$$

then the implied diffusion predictor has a strictly smaller Jacobian bound than a direct regressor with the same Lipschitz budget L .

Proof. The reconstruction formula is affine in the score field. For fixed y_t and t ,

$$J_c \hat{f}_{\theta,t}(c; y_t) = \frac{\sigma_t^2 + \alpha_t^2 \tau^2}{\alpha_t} J_c s_\theta(y_t, c, t). \quad (28)$$

Taking operator norms and applying the L -Lipschitz assumption gives the claim. $\square$

3.1 OOD implication

For any OOD point $c' \in \mathcal{S}_r$, let $c_0 \in \mathcal{S}$ be a nearest in-support point. Then

$$\|\hat{f}_{\theta,t}(c'; y_t) - \hat{f}_{\theta,t}(c_0; y_t)\| \leq \kappa_t L \|c' - c_0\|, \quad (29)$$

where

$$\kappa_t =$$

$\text{rac} \sigma_t^2 + \alpha_t^2 \tau^2 \alpha_t$. (30) Thus diffusion does not identify the correct OOD target, but it limits how fast the implied clean prediction can move away from the in-support prediction.

4 Comparison to Direct Regression

For direct regression,

$$g_\theta(c) = \text{network}(c), \quad (31)$$

and if the network is L -Lipschitz, then

$$\|J_c g_\theta(c)\| \leq L. \quad (32)$$

For score-based diffusion,

$$\hat{f}_{\theta,t}(c; y_t) = \frac{1}{\alpha_t} (y_t + \sigma_t^2 s_\theta(y_t, c, t)), \quad (33)$$

so

$$\|J_c \hat{f}_{\theta,t}(c; y_t)\| \leq \frac{\sigma_t^2}{\alpha_t} L. \quad (34)$$

Therefore, under equal Lipschitz budgets,

$$\text{direct regression:} \quad \|J_c g_\theta\| \leq L, \quad (35)$$

$$\text{score diffusion:} \quad \|J_c \hat{f}_{\theta,t}\| \leq \kappa_t L. \quad (36)$$

When $\kappa_t < 1$, diffusion imposes a stricter Jacobian bound.

This is the formal sense in which diffusion can give “better OOD Jacobian regularization.” The regularization is not due to magical OOD inference. It follows from a representational bottleneck: the network must represent $f(c)$ through a field whose sensitivity to c is bounded, and the reconstruction of $f(c)$ multiplies this sensitivity by κ_t .

5 Alternative Diffusion Parameterizations

The coefficient depends on the prediction target.

5.1 Score prediction

For score prediction,

$$s_\theta(y_t, c, t) \approx -\frac{y_t - \alpha_t f(c)}{\sigma_t^2}, \quad (37)$$

and

$$\hat{f}_{\theta,t} = \frac{y_t + \sigma_t^2 s_\theta}{\alpha_t}. \quad (38)$$

Thus

$$\boxed{\kappa_t^{\text{score}} = \frac{\sigma_t^2}{\alpha_t}}. \quad (39)$$

5.2 Noise prediction

If the model predicts noise,

$$\epsilon_\theta(y_t, c, t) \approx \epsilon, \quad (40)$$

then

$$y_t = \alpha_t f(c) + \sigma_t \epsilon, \quad (41)$$

so

$$\hat{f}_{\theta,t} = \frac{y_t - \sigma_t \epsilon_\theta(y_t, c, t)}{\alpha_t}. \quad (42)$$

Therefore,

$$J_c \hat{f}_{\theta,t} = -\frac{\sigma_t}{\alpha_t} J_c \epsilon_\theta, \quad (43)$$

and

$$\boxed{\kappa_t^\epsilon = \frac{\sigma_t}{\alpha_t}.} \quad (44)$$

5.3 Clean-data prediction

If the network directly predicts x_0 or y_0 ,

$$x_\theta(y_t, c, t) \approx f(c), \quad (45)$$

then there is no automatic schedule-induced shrinkage:

$$\boxed{\kappa_t^{x_0} = 1.} \quad (46)$$

Thus the implicit Jacobian regularization is strongest when the clean map is represented indirectly through a score, noise, or velocity field rather than directly as an x_0 -predictor.

6 Flow Matching: General Linear Interpolants

Now consider a flow-matching path from noise z to data $y_1 = f(c)$:

$$y_t = a_t y_1 + b_t z, \quad t \in [0, 1], \quad (47)$$

where $t = 0$ corresponds to noise and $t = 1$ corresponds to data.

The target velocity is

$$u_t = \dot{a}_t y_1 + \dot{b}_t z. \quad (48)$$

A flow-matching model learns

$$v_\theta(y_t, c, t) \approx u_t. \quad (49)$$

Given y_t and v_θ , we can solve the linear system

$$\begin{bmatrix} y_t \\ v_\theta \end{bmatrix} = \begin{bmatrix} a_t & b_t \\ \dot{a}_t & \dot{b}_t \end{bmatrix} \begin{bmatrix} y_1 \\ z \end{bmatrix}. \quad (50)$$

The recovered clean output is

$$\hat{y}_1 = \frac{\dot{b}_t y_t - b_t v_\theta(y_t, c, t)}{a_t \dot{b}_t - \dot{a}_t b_t}. \quad (51)$$

Differentiating with respect to c , holding y_t fixed,

$$J_c \hat{y}_1 = -\frac{b_t}{a_t \dot{b}_t - \dot{a}_t b_t} J_c v_\theta(y_t, c, t). \quad (52)$$

Thus, if v_θ is L -Lipschitz in c ,

$$\boxed{\|J_c \hat{f}_{\theta,t}\| \leq \left| \frac{b_t}{a_t \dot{b}_t - \dot{a}_t b_t} \right| L.} \quad (53)$$

Define the flow-matching reconstruction coefficient

$$\kappa_t^{\text{FM}} = \left| \frac{b_t}{a_t \dot{b}_t - \dot{a}_t b_t} \right|. \quad (54)$$

This is the direct analog of σ_t^2/α_t for score-based diffusion.

Theorem 2 (Flow-matching Jacobian coefficient). *Let $y_t = a_t y_1 + b_t z$ with $y_1 = f(c)$, and assume the velocity model $v_\theta(y_t, c, t)$ is L -Lipschitz in c for fixed (y_t, t) . If $a_t \dot{b}_t - \dot{a}_t b_t \neq 0$, then the clean predictor reconstructed from (y_t, v_θ) satisfies*

$$\|J_c \hat{f}_{\theta,t}\|_{\text{op}} \leq \kappa_t^{\text{FM}} L, \quad (55)$$

where

$$\kappa_t^{\text{FM}} = \left| \frac{b_t}{a_t \dot{b}_t - \dot{a}_t b_t} \right|. \quad (56)$$

7 Schedule Analysis

7.1 Rectified flow / OT path

For the standard rectified-flow path,

$$y_t = (1-t)z + ty_1. \quad (57)$$

Thus

$$a_t = t, \quad b_t = 1-t, \quad (58)$$

and

$$\dot{a}_t = 1, \quad \dot{b}_t = -1. \quad (59)$$

The determinant term is

$$a_t \dot{b}_t - \dot{a}_t b_t = t(-1) - 1(1-t) = -1. \quad (60)$$

Therefore,

$$\kappa_t^{\text{RF}} = 1-t. \quad (61)$$

So rectified flow gives

$$\|J_c \hat{f}_{\theta,t}\| \leq (1-t)L. \quad (62)$$

For any $t > 0$, this is strictly smaller than the direct-regression bound L . If $t \sim \text{Unif}[0, 1]$, the average coefficient is

$$\mathbb{E}[1-t] = \frac{1}{2}. \quad (63)$$

Thus, in this idealized analysis, uniformly trained rectified flow imposes an average Jacobian shrinkage factor of $1/2$ on the reconstructed clean predictor.

7.2 Trigonometric variance-preserving flow path

Consider the trigonometric path

$$a_t = \sin\left(\frac{\pi t}{2}\right), \quad b_t = \cos\left(\frac{\pi t}{2}\right). \quad (64)$$

Then

$$\dot{a}_t = \frac{\pi}{2} \cos\left(\frac{\pi t}{2}\right), \quad \dot{b}_t = -\frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right). \quad (65)$$

The determinant is

$$a_t \dot{b}_t - \dot{a}_t b_t = -\frac{\pi}{2} \left[\sin^2 \left(\frac{\pi t}{2} \right) + \cos^2 \left(\frac{\pi t}{2} \right) \right] \quad (66)$$

$$= -\frac{\pi}{2}. \quad (67)$$

Therefore,

$$\boxed{\kappa_t^{\text{trig}} = \frac{2}{\pi} \cos \left(\frac{\pi t}{2} \right)}. \quad (68)$$

This coefficient is always at most

$$\frac{2}{\pi} \approx 0.637. \quad (69)$$

Its average value under $t \sim \text{Unif}[0, 1]$ is

$$\int_0^1 \frac{2}{\pi} \cos \left(\frac{\pi t}{2} \right) dt = \frac{4}{\pi^2} \approx 0.405. \quad (70)$$

Thus the trigonometric path gives an even stronger average Jacobian shrinkage coefficient than the rectified path in this reconstruction analysis.

7.3 VP diffusion score schedule

For a variance-preserving diffusion schedule,

$$\alpha_t^2 + \sigma_t^2 = 1. \quad (71)$$

For score prediction,

$$\kappa_t^{\text{score}} = \frac{\sigma_t^2}{\alpha_t}. \quad (72)$$

This is less than one when

$$\sigma_t^2 < \alpha_t. \quad (73)$$

Using $\sigma_t^2 = 1 - \alpha_t^2$, this becomes

$$1 - \alpha_t^2 < \alpha_t, \quad (74)$$

or

$$\alpha_t^2 + \alpha_t - 1 > 0. \quad (75)$$

Thus

$$\boxed{\alpha_t > \frac{\sqrt{5} - 1}{2} \approx 0.618}. \quad (76)$$

Equivalently, in terms of signal-to-noise ratio

$$\text{SNR}_t = \frac{\alpha_t^2}{\sigma_t^2}, \quad (77)$$

the contraction regime is approximately

$$\text{SNR}_t > 0.618. \quad (78)$$

For noise prediction,

$$\kappa_t^\epsilon = \frac{\sigma_t}{\alpha_t}, \quad (79)$$

which is less than one iff

$$\alpha_t > \sigma_t, \quad (80)$$

or

$$\text{SNR}_t > 1. \quad (81)$$

Thus score prediction has a larger contraction region than noise prediction under a VP schedule.

7.4 VE score schedule

For a variance-exploding-style corruption,

$$y_t = y + \sigma_t z, \quad (82)$$

we can view $\alpha_t = 1$. For score prediction,

$$\kappa_t^{\text{VE}} = \sigma_t^2. \quad (83)$$

Thus the implied clean map is more Jacobian-regularized than direct regression when

$$\sigma_t^2 < 1. \quad (84)$$

The strength of the effect depends on the data scaling. If outputs are normalized to unit variance, small and moderate noise levels give contraction, while very large noise levels do not.

7.5 Summary table

Model / parameterization	Coefficient κ_t	Contraction condition
Direct regression	1	none
Score diffusion	σ_t^2 / α_t	$\sigma_t^2 < \alpha_t$
Noise-prediction diffusion	σ_t / α_t	$\sigma_t < \alpha_t$
VE score diffusion	σ_t^2	$\sigma_t^2 < 1$
Rectified flow	$1 - t$	$t > 0$
Trigonometric FM	$(2/\pi) \cos(\pi t/2)$	always < 1

Table 1: Schedule-dependent Jacobian coefficients for reconstructed clean predictors. Under a field Lipschitz bound L , the clean predictor has Jacobian bound $\kappa_t L$.

The important qualitative difference is that rectified flow and trigonometric flow matching have coefficients below one for almost all nontrivial times. VP diffusion has both contraction and expansion regimes depending on the noise level and prediction parameterization.

8 Why This Matters for OOD Generalization

OOD generalization is impossible to guarantee without assumptions. If the training support is $c \in [-1, 1]$, many functions can agree on the support but differ outside it. Diffusion or flow matching cannot identify the true extension by itself.

However, under Lipschitz constraints, not all extensions are equally easy to represent. Direct regression with Lipschitz budget L permits any extension satisfying

$$\|J_c g_\theta\| \leq L. \quad (85)$$

A diffusion or flow model with the same field Lipschitz budget L permits only clean-map extensions satisfying

$$\|J_c \hat{f}_{\theta,t}\| \leq \kappa_t L. \quad (86)$$

When $\kappa_t < 1$, the admissible class of clean-map extensions is smaller.

Therefore, for near-support OOD points

$$c' \notin \mathcal{S}, \quad \text{dist}(c', \mathcal{S}) \leq r, \quad (87)$$

diffusion and flow matching favor predictions that move more slowly away from the in-support graph. This is the precise sense in which they provide better OOD Jacobian regularization.

This effect is especially relevant in the low-variance regime. When $p(y | c)$ is concentrated near $f(c)$, the learned score or velocity field must encode the location of the conditional ridge. If the conditional ridge bends rapidly with c , then the score or velocity field must also change rapidly with c . A Lipschitz-constrained field therefore penalizes high-Jacobian conditional ridges.

9 Interpretation as Conditional Density Geometry

For low-variance conditionals,

$$p(y | c) \approx \mathcal{N}(f(c), \tau^2 I). \quad (88)$$

The high-density region is a thin tube around the graph of f . The Jacobian

$$J_c f(c) \quad (89)$$

is the tangent geometry of this tube. The score field describes how points away from the tube are pulled back toward it. Diffusion and flow matching do not merely fit the centerline; they fit a field around the centerline.

Thus, a rapidly changing $f(c)$ requires a rapidly changing denoising or velocity field. If that field is Lipschitz-constrained, the learned conditional density geometry is biased toward lower-curvature and lower-Jacobian extensions.

This also explains why the effect is not equivalent to ordinary regression. Regression supervises the centerline. Diffusion and flow matching supervise vector fields around the centerline over many noise levels. The same clean function must be consistently reconstructible from these fields.

10 Limitations

First, this is a near-support OOD regularization result, not a far-OOD correctness result. If c is far from the training support, Lipschitz bounds become weak and semantic extrapolation remains underdetermined.

Second, the result assumes a meaningful Lipschitz constraint on the score or velocity field with respect to the conditioning input. This may come from spectral normalization, gradient penalties, bounded weights, architecture, optimizer bias, or explicit regularization. Without such a constraint, the field can simply increase its c -Jacobian to represent a high-Jacobian $f(c)$.

Third, the comparison assumes equal Lipschitz budgets between regression and diffusion/flow networks. In practice, score and velocity networks may have different architectures, normalizations, conditioning mechanisms, and input scalings.

Fourth, the result depends on parameterization. Score prediction, noise prediction, velocity prediction, and direct x_0 -prediction induce different coefficients. Direct clean prediction does not automatically enjoy the same schedule-induced Jacobian shrinkage.

Fifth, the derivative statements hold for fixed y_t and t . Differentiating along a data-generation path $y_t(c) = \alpha_t f(c) + \sigma_t z$ introduces additional terms. The fixed- y_t derivative is the natural object for studying the learned conditional field as a function of the conditioning input.

Sixth, the result is cleanest for low-variance unimodal conditionals. For highly multimodal $p(y | c)$, there may be no single function $f(c)$, and the appropriate object becomes the geometry of multiple modes or the full conditional transport field.

11 Experimental Predictions

This theory suggests several testable predictions.

1. For low-variance conditional tasks, score- or velocity-based diffusion models should exhibit smaller empirical input-output Jacobian norms than direct regressors with comparable architecture and explicit Lipschitz control.
2. The effect should be stronger for schedules with smaller average reconstruction coefficient $\mathbb{E}_t[\kappa_t]$. Rectified flow should show an average factor near $1/2$, while trigonometric flow paths should show a factor near $4/\pi^2 \approx 0.405$ in the idealized uniform- t setting.
3. VP diffusion should show schedule-dependent behavior: moderate-noise score prediction should regularize the Jacobian, while very high-noise regions may not.
4. OOD performance improvements should be largest for near-support OOD inputs, where Lipschitz propagation from the training support is meaningful.
5. Removing the Lipschitz constraint should weaken or eliminate the effect.

12 Conclusion

We presented a simple theoretical argument that diffusion and flow-matching objectives can implicitly regularize the Jacobian of a learned low-variance conditional function. The key is that the clean predictor is reconstructed from a learned score or velocity field through a schedule-dependent coefficient κ_t . Under equal Lipschitz budgets, the implied clean predictor has Jacobian bounded by $\kappa_t L$, while direct regression has bound L . For many common flow-matching paths, including rectified and trigonometric paths, $\kappa_t < 1$ for most or all nontrivial times. For score-based diffusion, $\kappa_t = \sigma_t^2 / \alpha_t$, giving contraction in small-to-moderate noise regimes.

This gives a precise sense in which diffusion and flow matching may provide better OOD Jacobian regularization than standard regression in low-variance conditional learning: they do not identify the true OOD function, but they restrict the class of Lipschitz-compatible extensions by forcing the conditional mean or mode to be encoded through a regularized denoising or transport field.

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